



sparse step (queries = text + hard): selector $s_i = \text{rk}(M) + \frac{1}{2}\text{rk}(B) + \text{rk}(\Delta) [+ \text{rk}(R)]$
Lever A dual blocks: image q/FF on k rows **Lever B** K/V frozen at anchor $\Rightarrow 12S_q D^2$
 merged output: $v = \text{scatter}(v^a, \mathcal{H}, v^{\text{hard}})$ est. MAC $0.49 \rightarrow \mathbf{0.24\times}$

budget tiers
 low: reuse ($r=0$)
 mid: dual+KV refresh
 high: KV-only refresh